



Topological Sort, Homework 10 - Solution

1 Directed Graphs

1. Let G be a DAG, prove that $G^\top$ is also a DAG.
2. Proof the following claim¹

Claim 1. Let G be a directed graph. Define the relation $v \sim u$ to hold if u and v are on the same strong connected component. Then:

- (a) $\sim$ is an equivalence relation.
- (b) Define the graph $G' = (V', E')$ such that V' are the equivalence class respecting to $\sim$ and $\{u, v\} \in E'$ if there is a directed path in G from a vertex in u into a vertex in v . Then G' is a DAG.

Solution.

1. Assume for the sake of contradiction that $G^\top$ contains a cycle $v \rightsquigarrow v$, and denote the cycle path by $v = v_1, v_2, \dots, v_k = v$, such that $(v_i, v_{i+1}) \in E^\top$. Thus, we have that $(v_{i+1}, v_i) \in E$. Therefore, $v = v_k, v_{k-1}, \dots, v_2, v_1 = v$ is a cycle in G , in contradiction to G being a DAG.
2. (a) Let $v, u, w \in V$ such that $v \rightsquigarrow u$ and $u \rightsquigarrow w$ trough the paths $v = v_1, v_2, \dots, v_k = u$ and $u = u_1, u_2, \dots, u_l = w$ then the path

$$v = v_1, v_2, \dots, v_k, u_2, \dots, u_l = w$$

is a path $v \rightsquigarrow w$. By the same chaining technique, we get that the paths $w \rightsquigarrow u, u \rightsquigarrow v$ induce a path $w \rightsquigarrow v$. Therefore, directed connectivity is an equivalence relation.

- (b) Suppose not. Namely, there is a strongly connected component C_1 in G such that its associated vertex in G' belongs to a cycle $C_1 \rightsquigarrow C_1$. Let $C_2 \neq C_1$ be a strongly connected component in G such that its associated vertex in G' is on that cycle. By using the same chaining technique as in (a)² we get that for any $v \in C_1$ and $u \in C_2$ there are

¹From the recitation.

²Here we need to chain a path in G that might go through many vertices of each connected component on the $C_1 \rightsquigarrow C_2$ path in G' .

directed paths $v \rightsquigarrow u$ and $u \rightsquigarrow v$. Thus, the extension $C_1 \cup C_2$ is also connected (in G), in contradiction to the maximal property of a connected component.

2 Topological Sort

1. Algorithm 1 presents a variation of topological sort where the final DFS iterates over the vertices in increasing post-order instead of decreasing as in the original version. Give an example of a **family** of directed graphs for which the algorithm fails to return the strongly connected components.

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1 pre, post  $\leftarrow$  DFS( $G^\top$ )
2  $V \leftarrow$  sorts  $V$  in order of post.
3  $\pi \leftarrow$  DFS( $G$ )
4 return  $\pi$ 

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Algorithm 1: Wrong topological sort.

Solution.

1. Let Λ be the all-to-all connected graph over 5 vertices (5 clique), and let G_n be a n -length chain of Λ s such that the i th Λ copy is connected to the $(i + 1)$ th copy through a directed edge. Figure 1 illustrates an instance of the graph family.

Notice that $G^\top$ is the reverse chain, Namely, in the transpose graph, the i th copy is connected to the $i - 1$ th copy via a directed chain. So, the vertex with the minimal post value regarding DFS($G^\top$) is the first copy.

Thus, the first call to 'Explore' on the graph G is over a vertex of the first copy (leftmost copy in the figure), and since there is a path from it to any other vertex in G_n , all the vertices will be returned.

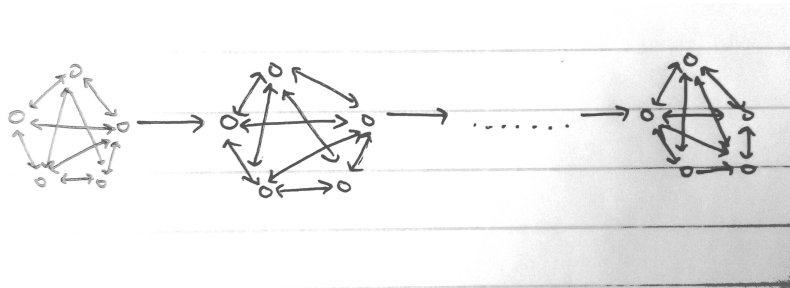


Figure 1: Example of family G_n , a n -length linear chain of cliques